HW 2

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Problems from: https://www.math.cmu.edu/~ellenp/HWK/ma355hw2.pdf
Due: 5/6/26

Problem 1

Let A, B, C be sets and let $f : A \longrightarrow B$, $g : B \longrightarrow C$ be functions. Prove: If f is onto B and g is onto C, then $g \circ f : A \longrightarrow C$ is onto C.

Solution

Since f is surjective, we have $\forall b \in B$, $\exists a \in A$ such that $f(a) = b$.

Since g is surjective, we have $\forall c \in C$, $\exists b \in B$ such that $f(b) = c$.

Let $c_0 \in C$ be arbitrary then $\exists b_0 \in b$ such that $g(b_0) = c_0$, by the second statement. Similarly $\exists a_0 \in a$ such that $f(a_0) = b_0$, by the first statement. So $g(f(a_0)) = c_0$. So we have shown that for any $c_0 \in C$, $\exists a_0 \in A$ such that $g(f(a_0)) = c_0$, and so $g \circ f$ is surjective. ■

Problem 2

Show that the relation $\sim$ (two sets are equivalent) is an equivalence relation.

Solution

We need three properties. Reflexivity, symmetry and transitivity. Let A, B, C be sets.

Trivially $A \sim A$. So $\sim$ is reflexive.

Clearly if A and B are the same set then $A \sim B$ and $B \sim A$. So $\sim$ is symmetric.

If A and B are the same set and B and C are the same set then A and C are the same set, so it means that $A \sim B$ and $B \sim C \implies A \sim C$. So $\sim$ is transistive. ■

Problem 3

Give an example of a countable collection of finite sets whose union is not finite.

Solution

For example the collection of sets

$$\{1\}, \{2\}, \{3\}, \ldots$$

is countable and it's union is $\mathbb{N}$ which is infinite. ■

Problem 4

Are the following sets finite, countable or uncountable? Explain or prove your answer in each case

1. $\{(x, y) \in \mathbb{N} \times \mathbb{R} : xy = 1\}$
2. $(\frac{1}{4}, \frac{3}{4})$

Solution

1. Note that y is uniquely determined by x so the set is equivalent to

$$\left\{\left(x, \frac{1}{x}\right) : x \in \mathbb{N}\right\}$$

which is clearly countable.

2. This interval is uncountable. Let us show it by contradiction. Suppose that f is a bijection mapping $\mathbb{N} \to (\frac{1}{4}, \frac{3}{4})$. We may assume that f is strictly increasing, since we can re-order the elements in the image of f as we wish. Define, for some fixed $k \in \mathbb{N}$

$$d = \frac{1}{2}\left(f(k+1) + f(k)\right).$$

We have $f(k) < d < f(k+1)$ so $d \in (\frac{1}{4}, \frac{3}{4})$ but d isn't in the image of f, a contradiction. ■

Problem 5

Is the set of all irrational numbers countable? Prove your answer.

Solution

The answer is no, the set of irrational numbers is uncountable. Define I to be the set of irrational numbers. Suppose FTSOC that I is countable. Then any non-empty subset of I is countable.

Let $a < b$ be real. Take $S(a, b) := \{x : x \in I,\ a \leq x \leq b\}$. Let $R(a, b) := \{x : x \in \mathbb{Q},\ a \leq x \leq b\}$. It is well-known that $\mathbb{Q}$ is countable and since $R(a, b) \subset \mathbb{Q}$, $R(a, b)$ is countable. Observe that

$$S(a, b) \cup R(a, b) = [a, b]$$

is countable since the union of countable sets is countable. However any non-empty interval of real numbers is uncountable, contradiction. ■

Problem 6

Prove that $\mathbb{N} \times \mathbb{N}$ is countable.
Hint: Consider $f(m, n) = 2^{m-1}(2n - 1)$.

Solution

The idea is to show that $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ is a bijection.

Note that for any $n \in \mathbb{N}$ we can write

$$n = 2^{v_2(n)} \frac{n}{2^{v_2(n)}}$$

where v_p is the p-adic valuation. We mean to say that any natural number can be expressed as a pure power of 2 multiplied by some odd number. Since $m \geq 1$, 2^{m-1} can be any power of 2 in $\mathbb{N}$, and $2n - 1$ can be any odd number in $\mathbb{N}$. Hence f is surjective.

Next we show that f is injective. Suppose that $f(a, b) = f(m, n)$. Clearly $a = m$ because otherwise $v_2(f(a, b)) \neq v_2(f(m, n))$. Thus we have

$$2n - 1 = 2b - 1 \iff n = b$$

and so f is injective. Hence $\mathbb{N} \times \mathbb{N}$ is in bijection with $\mathbb{N}$ and so it is countable. ■